Oxygen and sulfur appear to define a shared local geometry in multivariable periodic table space

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Table of Contents

1. **Abstract**
2. **Introduction**
3. **Variables and Methods**
4. **Main oxygen-centered geometry**
5. **Robustness and sensitivity**
6. **Alternative anchors**
7. **Oxygen versus sulfur**
8. **Limitations**
9. **Conclusion**
10. **References**

1. Abstract

We constructed a multivariable representation of the 118-element periodic table using electronegativity, electron affinity, first ionization energy, oxygen-affinity proxy, redox-flexibility proxy, group, and period. Principal component analysis was used to generate shared 2D and 3D spaces, and all elements were ranked by Euclidean distance to oxygen.

The oxygen-centered neighborhood did not appear random. Elements nearest to oxygen were consistently enriched in sulfur-family elements, halogens, selenium-related chemistry, and catalytically active transition metals, whereas alkali metals and noble gases remained distant.

This organization remained broadly stable across 2D and 3D embeddings, after removal of group and period, and within reduced high-confidence subsets. Among all tested alternative anchors, sulfur reproduced the oxygen top-10 neighborhood under the tested representation and generated the most compact local geometry.

These results suggest that oxygen may occupy a robust multivariable regime rather than an isolated position in chemical space, and that oxygen and sulfur may define a broader shared chemical axis characterized by redox flexibility, catalytic behavior, and cross-regime connectivity.

2. Introduction

Oxygen occupies a distinctive position in chemistry because it participates simultaneously in structural bonding, redox exchange, catalysis, and biological energy transfer. However, most conventional descriptions of oxygen rely on isolated variables such as electronegativity, oxidation state, or periodic-table position. These descriptors are useful, but they may not fully capture the broader multivariable neighborhood in which oxygen operates.

Here, we explored whether oxygen occupies a non-random local geometry when the full periodic table is embedded using multiple chemical properties simultaneously. We constructed a multivariable representation of all 118 elements using electronegativity, electron affinity, first ionization energy, oxygen-affinity proxy, redox-flexibility proxy, group, and period. Elements were then projected into shared 2D and 3D PCA spaces and ranked according to their Euclidean distance from oxygen.

The main objective was not to determine whether oxygen is universally "central," but whether it occupies a recognizable local neighborhood enriched in specific classes of chemically similar elements. We also tested whether this neighborhood remained stable under changes in dimensionality, removal of periodic-table variables, and restriction to high-confidence subsets.

Finally, we compared oxygen-centered geometry with alternative reference elements including carbon, nitrogen, hydrogen, and sulfur. This allowed us to evaluate whether oxygen occupies a uniquely broad multivariable regime or whether similar local structures also emerge around other chemically important anchors.

Overall, the analysis suggests that oxygen may occupy a robust local neighborhood enriched in sulfur-family elements, halogens, and catalytically active metals, while sulfur may represent the closest tested analogue to oxygen among the reference elements explored here.

3. Variables and Methods

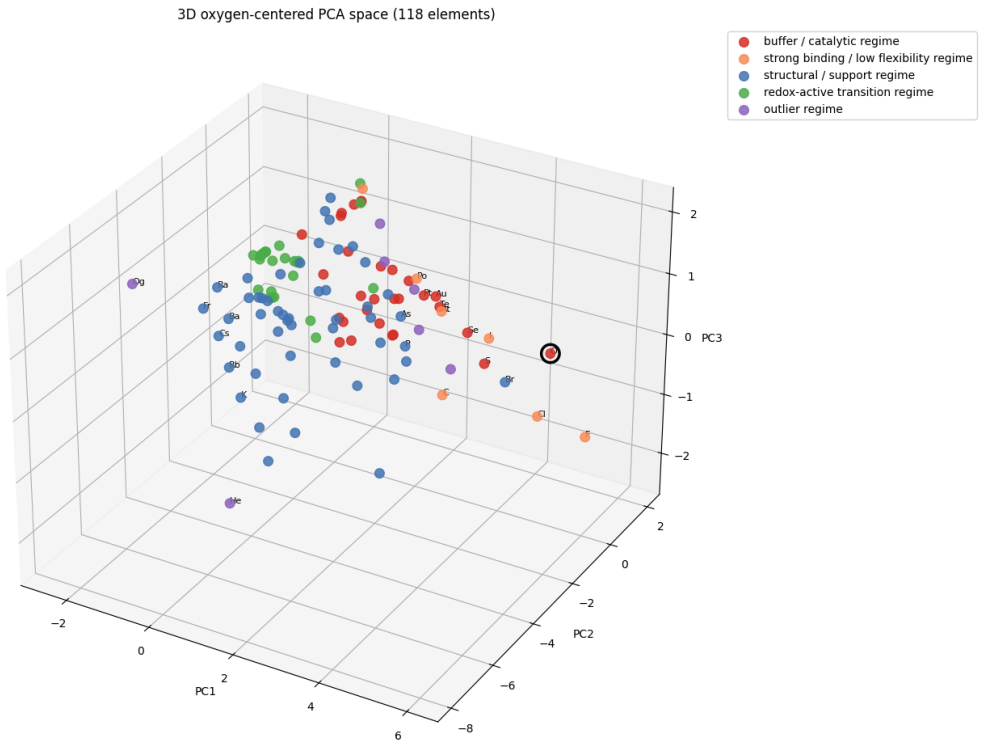

Figure 1: Elements are embedded in a three-dimensional PCA space constructed from electronegativity, electron affinity, first ionization energy, oxygen-affinity proxy, redox-flexibility proxy, group, and period. Oxygen is highlighted as the reference element. Nearby regions are enriched in sulfur-family elements, halogens, catalytic metals, and strong-binding elements, whereas alkali metals and noble gases occupy more distant regions

The analysis used a 118-element periodic table representation constructed from a set of continuous and categorical chemical-property variables.

The main variables included:

- electronegativity
- electron affinity
- first ionization energy
- oxygen-affinity proxy
- redox-flexibility proxy
- periodic-table group
- periodic-table period

All variables were standardized before dimensional reduction.

The oxygen-affinity proxy was constructed as an approximate measure of the tendency of each element to form stable oxygen-containing compounds.

The redox-flexibility proxy was constructed as an approximate measure of the diversity of accessible oxidation states and electron-exchange behavior across elements.

Two-dimensional and three-dimensional principal component analysis (PCA) embeddings were then generated to construct a shared multivariable chemical space. Distances between elements were calculated using Euclidean distance in both 2D and 3D PCA space.

Oxygen-centered rankings were constructed by measuring the distance of every element relative to oxygen. Additional rankings were generated around carbon, nitrogen, hydrogen, and sulfur in order to compare local neighborhood structure across alternative reference elements.

Several robustness analyses were performed. These included comparison between 2D and 3D embeddings, removal of categorical periodic-table variables, reduced high-confidence subsets, and comparison across alternative anchors.

The resulting framework does not attempt to establish causality or chemical mechanism. Instead, it provides a geometric representation of how elements cluster around oxygen and related reference elements in a shared multivariable space.

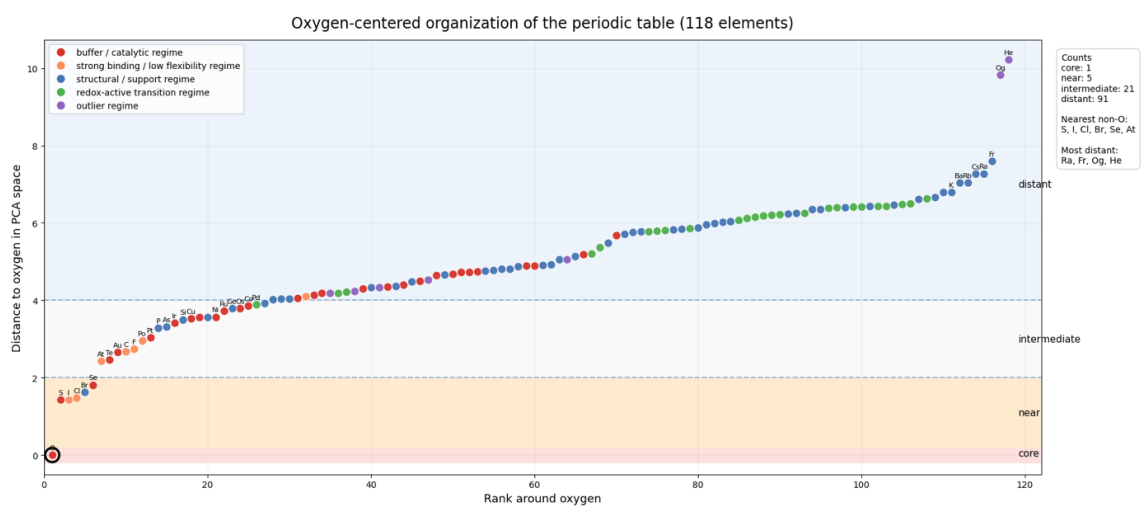

Figure 2: Elements are ranked according to their Euclidean distance from oxygen in the shared PCA space. The nearest region is dominated by sulfur, halogens, selenium-related chemistry, carbon, gold, and catalytic metals, whereas the most distant region is enriched in alkali metals, heavy structural elements, and noble-gas outliers. Colors indicate the assigned regime classification for each element.

4. Main oxygen-centered geometry

The oxygen-centered space did not appear random.

Elements nearest to oxygen were consistently enriched in sulfur-family elements, halogens, selenium-related chemistry, and catalytically active transition metals. As distance increased, the space progressively shifted toward structural metals, alkali elements, lanthanides, actinides, and noble-gas outliers.

This organization suggests that oxygen does not sit at the center of a generic “reactive” space, but rather inside a more specific multivariable region defined by strong electronegativity, substantial electron affinity, bond rigidity, and redox flexibility.

Catalytic and transition metals remained connected to oxygen-centered space, but generally through an intermediate shell rather than through direct proximity. In contrast, noble gases and alkali metals remained consistently distant across multiple representations.

The resulting geometry appears consistent with the idea that oxygen occupies a distinctive multivariable regime: not fully isolated, but surrounded by a limited set of chemically similar neighbors within the modeled space.

5. Robustness and sensitivity

To evaluate whether the oxygen-centered geometry depended strongly on a particular modeling choice, elemental rankings were compared across multiple model variants, including 2D PCA, 3D PCA, a chemical-only model excluding group and period, and a reduced high-confidence subset.

Most elements remained relatively stable across these alternative representations. The nearest-neighbor region around oxygen remained enriched in sulfur-family elements, halogens, selenium-related chemistry, and catalytically active metals across all tested variants. This suggests that the oxygen-centered structure is not simply an artifact of a single PCA configuration.

At the same time, a smaller subset of elements showed elevated sensitivity to dimensionality and variable selection. The most sensitive elements included astatine, fluorine, bromine, aluminum, chlorine, boron, tantalum, tungsten, lithium, and rhenium. These elements shifted substantially in rank depending on whether the model used 2D or 3D geometry, whether categorical periodic-table variables were removed, or whether the analysis was restricted to a high-confidence subset.

In contrast, oxygen itself, sulfur, carbon, tellurium, polonium, actinium, arsenic, radium, oganesson, and francium remained among the most stable elements across all tested configurations.

This suggests that the oxygen-centered geometry contains a robust structural core together with a smaller set of boundary elements whose position depends more strongly on the representation used. The strongest result is not that oxygen has nearby elements, but that its local neighborhood remains recognizable across multiple alternative model variants.

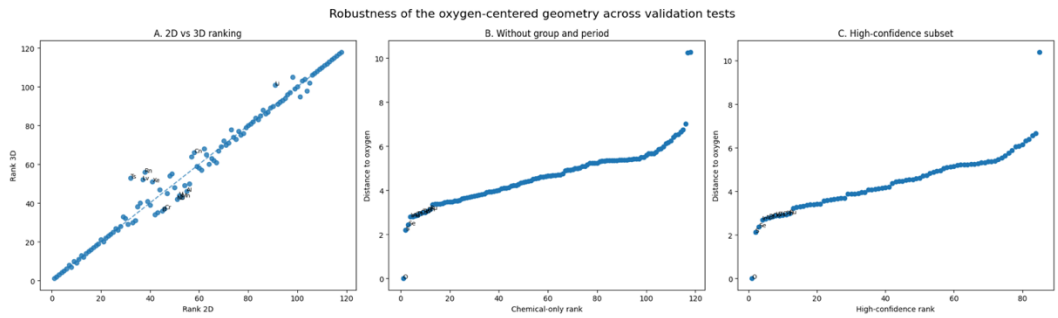

Figure 3: The oxygen-centered organization was compared across three alternative model variants: (A) 2D versus 3D PCA ranking, (B) a chemical-only model excluding group and period, and (C) a reduced high-confidence subset. Most elements remain close to the diagonal or preserve similar local structure across variants, suggesting that the oxygen-centered geometry is not driven by a single PCA configuration or by periodic-table ordering alone.

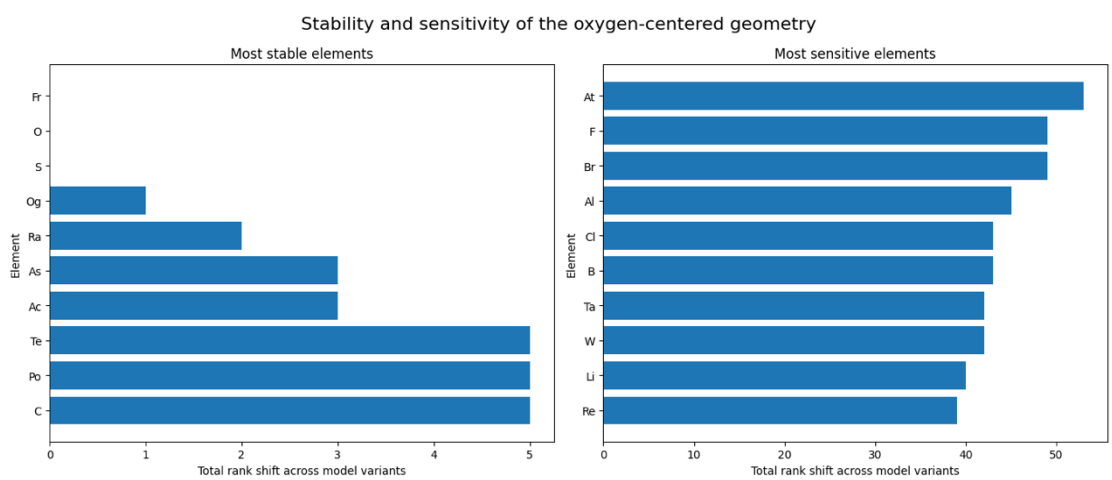

Figure 4: Elements were ranked according to the total shift in their oxygen-centered position across multiple model variants. The left panel shows the most stable elements, which remain in similar positions regardless of dimensionality or feature selection. The right panel shows the most sensitive elements, whose ranking changes substantially depending on model representation. These results suggest the presence of a stable structural core together with a smaller boundary region that is more representation-dependent.

6. Alternative anchors

To determine whether oxygen generated a uniquely structured neighborhood, additional rankings were constructed around carbon, nitrogen, hydrogen, and sulfur.

Carbon generated a more compact local neighborhood than oxygen, suggesting a tighter and more homogeneous structural chemistry. However, its local region showed lower overlap with the oxygen-centered top-10 set and appeared more strongly associated with structural and bonding-related properties.

Nitrogen and hydrogen generated substantially different local neighborhoods and showed relatively low overlap with the oxygen-centered space. Although both elements are chemically important, neither reproduced the same multivariable organization observed around oxygen.

Sulfur emerged as the closest tested analogue to oxygen among the evaluated reference elements. Sulfur reproduced the oxygen top-10 neighborhood under the tested representation and generated the most compact local geometry. Sulfur also showed the highest overlap with oxygen-centered rankings across the broader top-20 region.

These results suggest that oxygen-centered organization is not merely a generic feature of any chemically important element. Instead, oxygen appears to belong to a more specific oxygen-sulfur family characterized by redox flexibility, catalytic behavior, and strong cross-regime connectivity.

7. Oxygen versus sulfur

The comparison between oxygen and sulfur produced the strongest result of the present analysis.

Sulfur did not merely appear near oxygen in multivariable space. Instead, sulfur reproduced the oxygen top-10 neighborhood under the tested representation while generating an even more compact local geometry.

The local neighborhoods around oxygen and sulfur contained the same nearest elements, including sulfur-family elements, halogens, selenium-related chemistry, and catalytically active metals. However, sulfur generated a tighter local cluster, whereas oxygen extended into a broader and more heterogeneous space.

This pattern may suggest that sulfur defines the most compact redox-centered local core, whereas oxygen may act as the broader bridge connecting multiple chemical regimes.

One possible interpretation is that oxygen and sulfur occupy a shared chemical axis characterized by redox flexibility, catalytic mediation, exchange capacity, and electronic adaptability. Under

this view, oxygen may represent the more expansive and heterogeneous end of the axis, while sulfur may represent the most compact and locally stable core within the present representation.

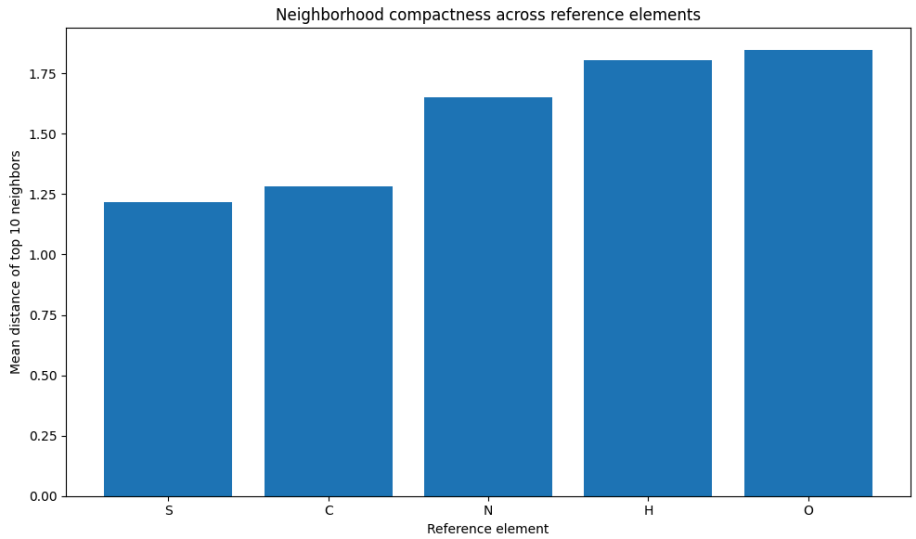

Figure 5: The mean distance of the top-10 nearest neighbors was compared across multiple reference elements. Sulfur generated the most compact local neighborhood, followed by carbon, whereas oxygen produced the broadest neighborhood among the tested reference elements. This suggests that sulfur occupies a more locally concentrated region of the same broader geometry.

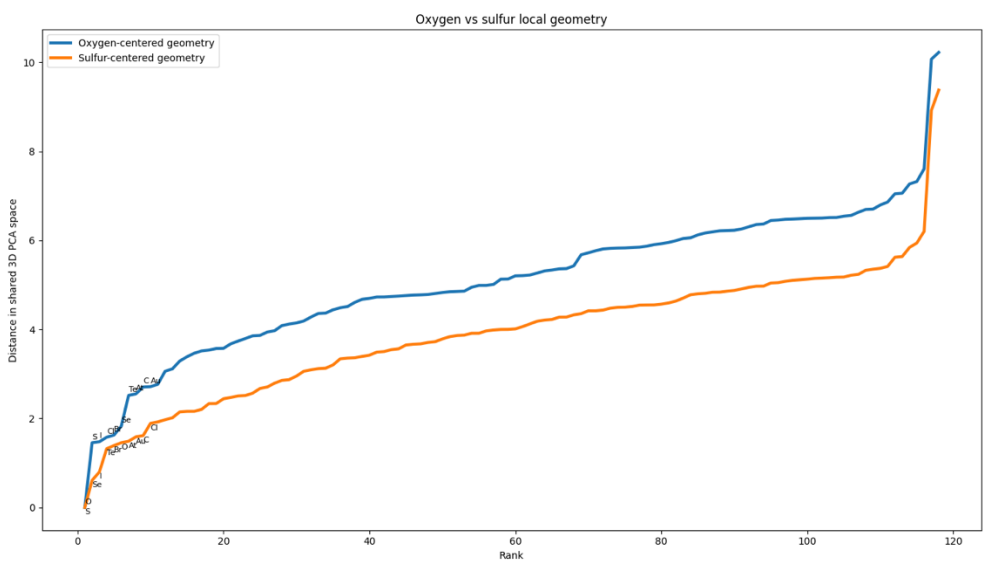

Figure 6: Rank-distance curves for oxygen-centered and sulfur-centered spaces are shown across all 118 elements. Sulfur reproduces the local oxygen neighborhood under the tested representation while maintaining a more compact overall geometry. This result suggests that oxygen and sulfur may define a shared multivariable chemical axis rather than two unrelated local structures.

8. Limitations

Several limitations should be considered when interpreting these results.

First, the geometry depends on the selected variables and on the use of PCA as a dimensional reduction method. Different variable choices or nonlinear embeddings could potentially alter the resulting neighborhoods.

Second, oxygen was explicitly chosen as one of the reference elements. The present analysis therefore does not demonstrate that oxygen is universally central in an absolute sense. Instead, it shows that when the periodic table is organized around oxygen, a robust and non-random local neighborhood emerges.

Third, some elements required estimated or proxy values, especially within the least experimentally constrained regions of the periodic table. Although high-confidence subset analyses produced similar results, some uncertainty remains for the most weakly characterized elements.

Finally, the present work does not establish chemical causality or mechanism. The observed organization should be interpreted as a geometric and multivariable clustering pattern rather than as direct evidence of specific reaction pathways or physical constraints.

9. Conclusion

The present analysis suggests that oxygen occupies a distinctive local neighborhood in multivariable periodic-table space. This neighborhood remains recognizable across alternative dimensionalities, reduced feature sets, and high-confidence subsets, suggesting that the observed organization is not simply an artifact of periodic-table ordering or a single PCA configuration.

The nearest elements to oxygen were consistently enriched in sulfur-family elements, halogens, selenium-related chemistry, and catalytically active metals, whereas noble gases and alkali metals remained distant.

Among all tested alternative anchors, sulfur emerged as the closest tested analogue to oxygen. Sulfur reproduced the oxygen top-10 neighborhood under the tested representation while generating a more compact local geometry. This suggests that oxygen may not act as an isolated center, but instead may belong to a broader oxygen-sulfur chemical family characterized by redox flexibility, catalytic behavior, and cross-regime connectivity.

More broadly, the distinction between stable local cores and representation-sensitive boundary regions suggests that the oxygen-centered space is not homogeneous.

Instead, it appears to contain robust neighborhoods together with more weakly constrained frontier regions.

10. References

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